

Practical 5 - COS 214

Livia Ami Webber u24607852

Ishmael Ngwasheng u24642542

Prince Tamiru u25270665

Task 1

Incident-Response Responsibilities

CampusGuard supports campus incident-response staff from the moment an incident is reported until it is resolved.

The system works as follows:

1. An incident is registered and their operational status is tracked through a defined lifecycle (reported, dispatched, in progress, resolved, or cancelled at any point before resolution)
2. An operator dispatches security, medical, or facilities response units, issues evacuation instructions, activates campus alerts, and cancels previously issued instructions
3. Response units coordinate without needing to hold direct references to one another
4. Integrates the campus's existing legacy building access-control system so it can be locked down or released through CampusGuard's own interface, without CampusGuard code depending on the legacy system's method names directly
5. exposes a single high-level operation for common emergency workflows (e.g. "report an emergency") that would otherwise require an operator to manually sequence several separate subsystem calls
6. Keeps a live dashboard and an audit log informed whenever an incident's status changes, without those stakeholders being able to instruct response units themselves

GoF Participants

Command

- Command: Command (interface)
- ConcreteCommand: DispatchUnitCommand, IssueEvacuationCommand, ActivateAlertCommand, CancelActionCommand
- Invoker: OperatorConsole
- Receiver: ResponseUnit and its subclasses

Mediator

- Mediator: IncidentMediator (interface)
- ConcreteMediator: IncidentCoordinator
- Colleague: ResponseUnit
- ConcreteColleague: SecurityTeam, MedicalTeam, FacilitiesTeam

Adapter

- Target: AccessControlService (interface)
- Adaptee: LegacyAccessControlSystem (external/legacy)
- Adapter: AccessControlAdapter

Facade

- Facade: EmergencyResponseFacade
- Subsystem classes: Incident, OperatorConsole/Command, IncidentCoordinator, AccessControlAdapter

Additional GoF Patterns Selected

State

- Context: Incident
- State: IncidentState (abstract)
- ConcreteState: ReportedState, DispatchedState, InProgressState, ResolvedState, CancelledState

Why State: An incident's valid operations genuinely change depending on where it is in its lifecycle. A Reported incident cannot be resolved directly, and you should not be able to dispatch units to a resolved one. This can be represented with a plain status field and then use if or switch everywhere a method reads or changes the status, but this is exactly the kind of type-checking the brief rules out for the patterns required. Additionally, any change to the lifecycle rules would mean you have to change every place that checks the status. The State pattern avoids this by internalising the transition logic within each life cycle stage. To add a new stage, you add a new class without changing any existing classes.

Observer

- Subject: Incident
- Observer: IncidentObserver (interface)
- ConcreteObserver: Dashboard, IncidentLogger

Why Observer: There are multiple independent parts of the system that need to react when the status of an incident changes: the live dashboard for staff, the audit log for compliance, and many others in a real deployment, such as SMS alerts and a campus-wide notice board. None of these should be hard-coded into Incident's logic or folded into the Mediator's active coordination of response units. The Mediator coordinates the units performing the response, while the Observer keeps the passive stakeholders informed.

Where the six patterns interact

EmergencyResponseFacade::reportEmergency() traverses all six patterns in one flow:

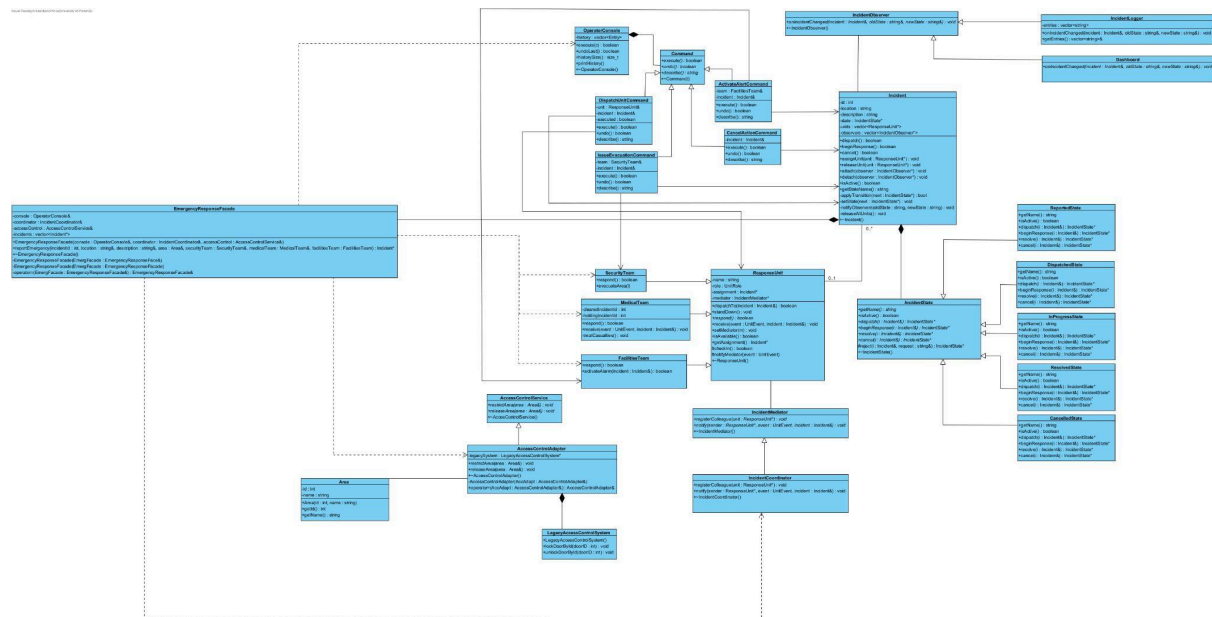
1. Facade creates an Incident, which starts life in the Reported State.
2. Facade submits a DispatchUnitCommand (Command) through OperatorConsole, which executes it against a SecurityTeam Receiver.
3. The Incident's state transitions to Dispatched, which triggers notifyObservers() (Observer) - the Dashboard and IncidentLogger pick this up independently of anything the Mediator or Command are doing.
4. Once SecurityTeam secures the area, it reports the event to IncidentCoordinator (Mediator), which coordinates MedicalTeam and FacilitiesTeam without any of the three teams knowing about each other directly.
5. Facade also calls AccessControlAdapter.restrictArea() (Adapter), which translates that call into the legacy system's own lockDoorByld() interface.

- When the incident is resolved or cancelled, setState() recognises it is no longer active and releases every assigned unit. State acts directly on the Mediator's colleagues.

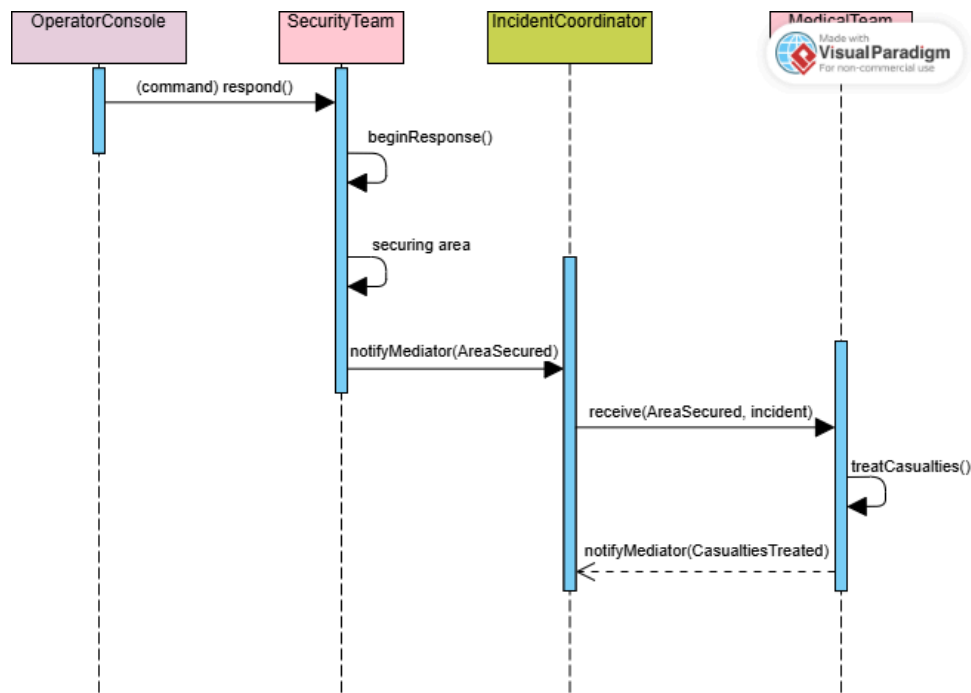
Single class participates in more than one pattern: ResponseUnit is simultaneously a Command Receiver and a Mediator Colleague (the same concrete team objects fulfill both roles). Incident serves as the Context in the State pattern and the Subject in the Observer pattern.

Task 4

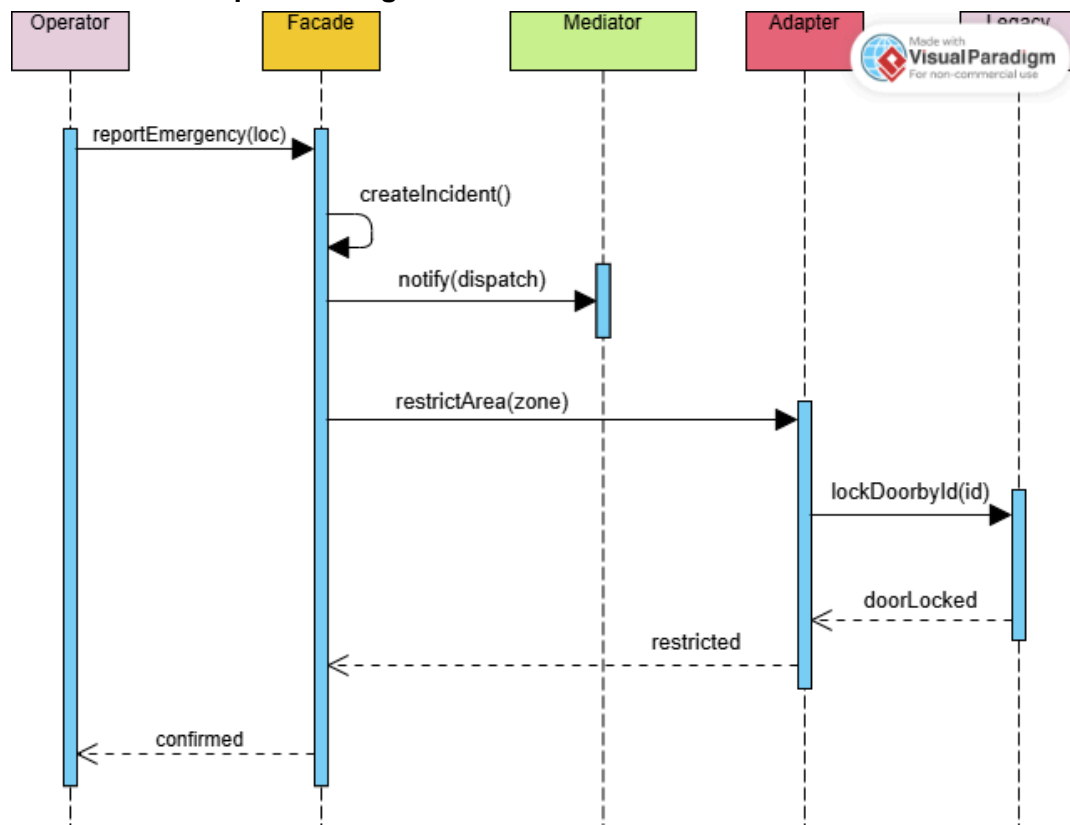
UML class diagram for the final system



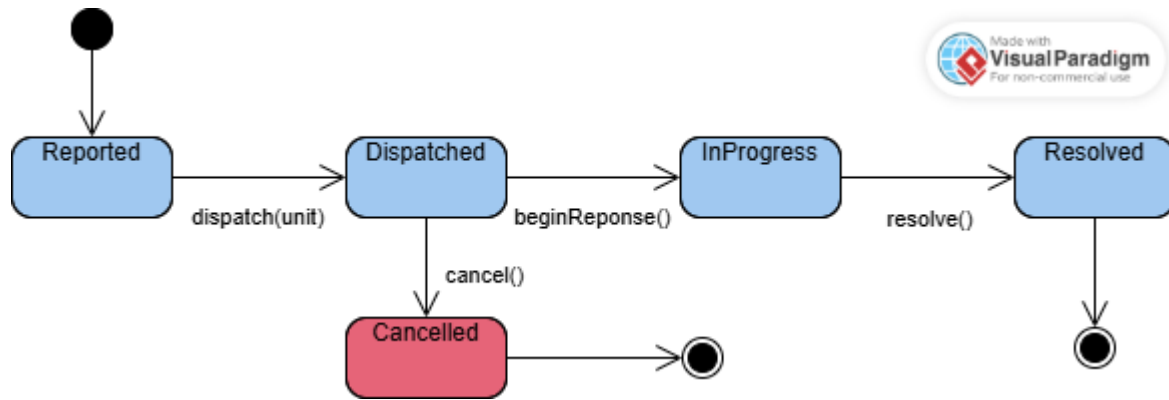
UML Command Sequence Diagram



UML Facade Sequence Diagram



State Diagram



Task 5

Valgrind Evidence & GDB Investigation

```
base) ishmael@ishmaelKaay:~/projects/214_PRACS/PRAC_5/COS214_Practical5$ make debug
```

```
LegacyAccessControlSystem] Locking door with ID: 202
Console] >> DispatchUnit(Campus Security -> Incident #1002)
Campus Security] dispatched to Incident #1002 (Residence Hall)
Console] OK
Campus Medical] dispatched to Incident #1002 (Residence Hall)
Campus Facilities] dispatched to Incident #1002 (Residence Hall)
Facade] Emergency response setup complete.
Console] >> CancelAction(Incident #1002)
Dashboard] Incident #1002 (Residence Hall): Dispatched -> Cancelled
Campus Security] stood down from Incident #1002
Campus Medical] stood down from Incident #1002
Campus Facilities] stood down from Incident #1002
Console] OK
Console] >> IssueEvacuation(Campus Security, Incident #1002)
Campus Security] is not assigned to Incident #1002
Console] FAILED
LegacyAccessControlSystem] Unlocking door with ID: 202
```

```
=== OPERATOR COMMAND HISTORY ===
```

```
Console] DispatchUnit(Campus Security -> Incident #1001) : OK
Console] IssueEvacuation(Campus Security, Incident #1001) : OK
Console] ActivateAlert(Campus Facilities, Incident #1001) : OK
Console] DispatchUnit(Campus Security -> Incident #1002) : OK
Console] CancelAction(Incident #1002) : OK
Console] IssueEvacuation(Campus Security, Incident #1002) : FAILED
```

```
=== INCIDENT AUDIT LOG ===
```

```
Logger] Incident #1001: Dispatched -> InProgress
Logger] Incident #1001: InProgress -> Resolved
Logger] Incident #1002: Dispatched -> Cancelled
```

```
Integration demo complete.
```

```
Inferior 1 (process 5256) exited normally]
```

```
(gdb) backtrace
```

```
to stack.
```

```
(gdb) █
```

```
base) ishmael@ishmaelKaay:~/projects/214_PRACS/PRAC_5/COS214_Practical5$ make debug
```

```
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/AccessControlAdapter.cpp -o build/AccessControlAdapter.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/Area.cpp -o build/Area.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/EmergencyResponseFacade.cpp -o build/EmergencyResponseFacade.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/LegacyAccessControlSystem.cpp -o build/LegacyAccessControlSystem.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/command/ActivateAlertCommand.cpp -o build/command/ActivateAlertCommand.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/command/CancelActionCommand.cpp -o build/command/CancelActionCommand.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/command/DispatchUnitCommand.cpp -o build/command/DispatchUnitCommand.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/command/IssueEvacuationCommand.cpp -o build/command/IssueEvacuationCommand.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/command/OperatorConsole.cpp -o build/command/OperatorConsole.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/domain/FacilitiesTeam.cpp -o build/domain/FacilitiesTeam.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/domain/Incident.cpp -o build/domain/Incident.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/domain/MedicalTeam.cpp -o build/domain/MedicalTeam.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/domain/ResponseUnit.cpp -o build/domain/ResponseUnit.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/domain/SecurityTeam.cpp -o build/domain/SecurityTeam.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/mediator/IncidentCoordinator.cpp -o build/mediator/IncidentCoordinator.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/observer/Dashboard.cpp -o build/observer/Dashboard.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/observer/IncidentLogger.cpp -o build/observer/IncidentLogger.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/state/CancelledState.cpp -o build/state/CancelledState.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/state/DispatchedState.cpp -o build/state/DispatchedState.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/state/InProgressState.cpp -o build/state/InProgressState.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/state/IncidentState.cpp -o build/state/IncidentState.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/state/ReportedState.cpp -o build/state/ReportedState.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c src/state/ResolvedState.cpp -o build/state/ResolvedState.o
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -c main.cpp -o build/main.o
```

```
++ -std=c++11 -Wall -Wextra -pedantic -g -Iinclude -o campusguard build/AccessControlAdapter.o build/Area.o build/EmergencyResponseFacade.o build/LegacyAccessControlSystem.o build/command/ActivateAlertCommand.o build/command/CancelActionCommand.o build/command/DispatchUnitCommand.o build/command/IssueEvacuationCommand.o build/command/OperatorConsole.o build/domain/FacilitiesTeam.o build/domain/Incident.o build/domain/MedicalTeam.o build/domain/ResponseUnit.o build/domain/SecurityTeam.o build/mediator/IncidentCoordinator.o build/observer/Dashboard.o build/observer/IncidentLogger.o build/state/CancelledState.o build/state/DispatchedState.o build/state/InProgressState.o build/state/IncidentState.o build/state/ReportedState.o build/state/ResolvedState.o build/main.o
```

```
gdb ./campusguard
```

```
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
```

```
Copyright (C) 2022 Free Software Foundation, Inc.
```

```
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
```

```
This is free software: you are free to change and redistribute it.
```

```
There is NO WARRANTY, to the extent permitted by law.
```

```
(base) ishmael@ishmaelkaay:~/projects/214_PRACS/PRAC_5/COS214_Practical5$ make valgrind
```

```
[Facade] Emergency response setup complete.
```

```
[console] >> CancelAction(Incident #1002)
```

```
[Dashboard] Incident #1002 (Residence Hall): Dispatched -> Cancelled
```

```
[Campus Security] stood down from Incident #1002
```

```
[Campus Medical] stood down from Incident #1002
```

```
[Campus Facilities] stood down from Incident #1002
```

```
[console] OK
```

```
[console] >> IssueEvacuation(Campus Security, Incident #1002)
```

```
[Campus Security] is not assigned to Incident #1002
```

```
[console] FAILED
```

```
[LegacyAccessControlSystem] Unlocking door with ID: 202
```

```
=== OPERATOR COMMAND HISTORY ===
```

```
[console] DispatchUnit(Campus Security -> Incident #1001) : OK
```

```
[console] IssueEvacuation(Campus Security, Incident #1001) : OK
```

```
[console] ActivateAlert(Campus Facilities, Incident #1001) : OK
```

```
[console] DispatchUnit(Campus Security -> Incident #1002) : OK
```

```
[console] CancelAction(Incident #1002) : OK
```

```
[console] IssueEvacuation(Campus Security, Incident #1002) : FAILED
```

```
=== INCIDENT AUDIT LOG ===
```

```
[logger] Incident #1001: Dispatched -> InProgress
```

```
[logger] Incident #1001: InProgress -> Resolved
```

```
[logger] Incident #1002: Dispatched -> Cancelled
```

```
=====  
Integration demo complete.  
=====
```

```
--3868==
```

```
--3868== HEAP SUMMARY:
```

```
--3868==    in use at exit: 0 bytes in 0 blocks
```

```
--3868== total heap usage: 83 allocs, 83 frees, 76,581 bytes allocated
```

```
--3868==
```

```
--3868== All heap blocks were freed -- no leaks are possible
```

```
--3868==
```

```
--3868== For lists of detected and suppressed errors, rerun with: -s
```

```
--3868== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```